

zkBSA: Auditable and Compliant Stealth Addresses for Blockchains

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Abstract. Stealth addresses provide receiver unlinkability on blockchains, but existing schemes do not support regulatory compliance in settings where transaction amounts are transparent and recipient identities should remain private. We present zkBSA, a modular framework for auditable and compliant stealth addresses. zkBSA combines a stealth address scheme, public-key encryption, a whitelist commitment, and zero-knowledge proofs to enable on-chain verification that a transaction targets a whitelisted receiver without revealing which receiver was chosen. At the same time, authorized auditors can recover receiver-linked audit information. We formalize security in terms of public-transcript receiver unlinkability, compliance soundness, and audit correctness, and show that these properties follow from standard assumptions on the underlying primitives. We implement a proof of concept using ERC-5564, EC-ElGamal, Merkle trees, and RISC Zero zkVM. Our evaluation shows proof generation under 5.3 seconds for whitelist sizes up to 2^{24} and a fixed on-chain verification cost of about 235k gas, indicating that zkBSA is practical for receiver-private, compliance-enforced blockchain transactions.

Keywords: Stealth Addresses · Receiver Unlinkability · Regulatory Compliance · Zero-Knowledge Proofs

1 Introduction

Blockchain systems offer transparency through publicly verifiable ledgers. Yet this transparency also exposes transaction metadata that can enable profiling and deanonymization [14]. Stealth addresses mitigate this problem by allowing recipients to receive funds through unlinkable one-time addresses [17], ensuring that external observers cannot link multiple payments to the same entity—a property known as receiver unlinkability. Stealth addresses have been applied in public blockchains such as Monero, Ethereum (through ERC-5564 [20]), as well as in consortium blockchain settings [9].

While highly effective for preserving user privacy, purely anonymous systems such as Monero and TornadoCash [15] are increasingly at odds with global regulatory frameworks. Regulations such as the FATF Travel Rule mandate that Virtual Asset Service Providers (VASPs) verify the identities of transacting parties and ensure transactions occur only between compliant entities [8]. This creates

a fundamental tension: strong privacy mechanisms that hide receiver identity from everyone are difficult to reconcile with compliance requirements demanding verifiable policy enforcement.

This tension is especially important in *amount-transparent but identity-sensitive* applications such as institutional treasury transfers, regulated stablecoin payments, and compliance-gated token ecosystems. In these settings, transaction amounts are often intentionally public or at least not the primary privacy concern, while the identities of counterparties should remain hidden from the public to reduce profiling, strategic inference, targeted attacks, and reputational leakage. This setting arises naturally in practice, yet it is not well served by existing approaches.

A particularly relevant deployment scenario arises at the boundary between regulated privacy pools and account-based blockchain systems. In regulated privacy systems or compliant mixers, such as Zeto [10], users may accumulate private UTXOs and later withdraw or consolidate funds into an account-based address for settlement or operational use. This exit step creates a privacy-compliance dilemma: withdrawing to a standard address exposes the destination publicly, whereas withdrawing to an unrestricted one-time address weakens receiver-side compliance enforcement. More broadly, such systems can protect privacy inside the pool, yet still lose recipient privacy at the exit boundary.

To this end, we propose zkBSA, a framework for auditable and compliant stealth addresses. Our model explicitly separates two orthogonal threats: (i) a privacy adversary (e.g., a passive public observer) who should not be able to determine which whitelisted receiver was paid from transaction data alone, and (ii) a compliance adversary (e.g., a malicious sender) who may attempt to bypass policy checks while producing valid transactions.

A key design choice is that the whitelist $\mathbf{W}$ of compliant receivers is public. Thus, our privacy goal is *not* to hide the set of compliant entities, but to prevent any public observer from linking an accepted transaction to a specific entry in $\mathbf{W}$.

In summary, our contributions are:

1. We propose **zkBSA, a modular framework for auditable, compliance-enforced stealth addresses**, which reconciles receiver unlinkability with enforceable compliance through a composition of standard cryptographic primitives, enabling flexible deployment across diverse blockchain platforms.
2. We define **a dual-objective security model and formal security properties**, capturing public-transcript receiver unlinkability against public observers, compliance soundness against malicious senders, and audit correctness for authorized auditors.
3. We implement and evaluate **a practical prototype on a realistic stack** (ERC-5564 + EC-ElGamal [7, 18] + Merkle trees + RISC Zero zkVM [3]), demonstrating the feasibility of achieving amount-transparent, receiver-private, and compliance-enforced transactions simultaneously in a real-world blockchain context.

Our work bridges privacy-enhancing technologies (PETs) and regulatory technology (RegTech), offering a framework for amount-transparent, receiver-private, and compliance-enforced blockchain transactions.

2 Related Works

Research on privacy-preserving blockchain transactions has evolved along several complementary paths, each addressing different parts of the privacy-compliance spectrum.

Stealth Address Formalization and Extensions. The theoretical underpinnings of stealth addresses have been significantly strengthened through formal cryptographic modeling. Backes et al. [1] provided the first rigorous abstraction, which formally captures public-key derivation, private-key recovery, and unlinkability. Subsequent work by Liu et al. [11] identified a practical security issue in naive implementations: compromise of a one-time spending key may leak the long-term master secret key. They introduced key insulation to mitigate this risk. This security model was further advanced by Pu et al. [16] with Spirit, a post-quantum secure construction that also features scalable fuzzy tracking. Recognizing the long-term threat posed by quantum computing, Mikić et al. [12] developed post-quantum alternatives to classical elliptic-curve-based stealth addresses.

Concurrently, efforts have been made to enhance the usability of stealth addresses in organizational contexts. Yin et al. [21, 22] formalized the concept of a Hierarchical Deterministic Wallet supporting Stealth Addresses (HDWSA). Their framework combines the privacy benefits of stealth addresses with the structured key management of HD wallets, enabling secure delegation within hierarchical entities while integrating signature aggregation. Other works have explored functional enhancements. Yu [23] introduced a scheme that allows a single wallet to manage multiple unlinkable addresses via child indices. Mongardini et al. [13] proposed an Identity-Based Matchmaking Stealth Signature (IB-MSS) scheme that embeds user attributes (e.g., KYC status) into the address-derivation process, where a transaction is spendable only if both parties' attributes satisfy policy requirements.

Confidential Payment Systems with Auditability. A distinct line of research focuses on confidential transactions (CT), where the goal is to hide transaction amounts [4]. Systems such as Anonymous Zether [6], PGC [5], and Anonymous PGC [24] operate within the standard account model and achieve receiver anonymity by leveraging homomorphic encryption over account balances. Transactions in these systems update encrypted balances of multiple accounts simultaneously, obscuring the true receiver. This powerful model provides comprehensive privacy but incurs significant overhead from on-chain homomorphic operations.

These confidential payment systems are designed for scenarios where both amount and identity confidentiality are paramount. In contrast, the stealth-address primitive is a natural fit for the identity-sensitive, amount-transparent setting common in regulated financial applications.

Positioning of our work. Existing stealth-address schemes have been rigorously formalized for security [1, 11, 16], extended to post-quantum settings [12, 16] and hierarchical structures [21, 22], and augmented with attribute-based policies [13]. However, they do not provide the combination targeted in this paper: *publicly verifiable on-chain whitelist enforcement together with receiver unlinkability and transaction-level auditor recovery of receiver-linked information in amount-transparent settings*. Unlike attribute-driven stealth-signature approaches such as IB-MSS [13], zkBSA provides publicly verifiable receiver-side whitelist enforcement tied to a committed compliance set, together with transaction-level auditor recovery of a receiver-linked public key and an account-model deployment path for amount-transparent blockchains. Our work addresses this gap by enabling preventive cryptographic enforcement of a compliance policy while preserving receiver unlinkability from the public.

3 Preliminaries

In this section, we introduce the cryptographic primitives and building blocks underlying our framework. In this paper, λ denotes the security parameter, $\text{poly}(\lambda)$ denotes any polynomial function in λ , and $\text{negl}(\lambda)$ denotes a negligible function.

3.1 Stealth Address Schemes

Stealth-address (SA) schemes enable receivers to generate unlinkable one-time addresses for receiving funds while maintaining the ability to spend them using their long-term secret keys. It consists of:

1. $\mathcal{SA}.\text{Setup}(1^\lambda) \rightarrow \text{pp}$: Initializes the public parameters pp .
2. $\mathcal{SA}.\text{SpendKeyGen}(\text{pp}) \rightarrow (\text{sk}_{\text{spend}}, \text{pk}_{\text{spend}})$: Generates a long-term spending key pair.
3. $\mathcal{SA}.\text{ViewKeyGen}(\text{pp}) \rightarrow (\text{sk}_{\text{view}}, \text{pk}_{\text{view}})$: Generates a long-term viewing key pair.
4. $\mathcal{SA}.\text{DeriveAddr}((\text{pk}_{\text{spend}}, \text{pk}_{\text{view}}), r) \rightarrow (\text{addr}, \text{hint})$: Given a receiver's public key pair and sender-chosen randomness r , derives a one-time stealth address addr and a public hint hint that enables the receiver to recognize the transaction during blockchain scanning.
5. $\mathcal{SA}.\text{CheckAddr}(\text{sk}_{\text{view}}, \text{hint}, \text{addr}) \rightarrow \{0, 1\}$: Allows the receiver to efficiently verify whether a given transaction with $(\text{hint}, \text{addr})$ is intended for them, using only their viewing private key sk_{view} .
6. $\mathcal{SA}.\text{RecoverKey}((\text{sk}_{\text{spend}}, \text{sk}_{\text{view}}), \text{hint}) \rightarrow \text{sk}_{\text{addr}}$: Recovers the private key sk_{addr} corresponding to the derived address addr .

Unlinkability. For any PPT adversary $\mathcal{A}$, the advantage in distinguishing whether two stealth addresses with hints $(\text{addr}, \text{hint}), (\text{addr}', \text{hint}')$ correspond to the same receiver or to different receivers is at most $\text{negl}(\lambda)$.

3.2 Public-Key Encryption

Our framework employs a public-key encryption (PKE) scheme to securely transmit audit information to authorized parties. The PKE scheme consists of:

1. $\mathcal{PKE}.\text{Setup}(1^\lambda) \rightarrow \text{pp}$: Initializes public parameters.
2. $\mathcal{PKE}.\text{KeyGen}(\text{pp}) \rightarrow (\text{sk}, \text{pk})$: Generates a secret/public key pair.
3. $\mathcal{PKE}.\text{Enc}(\text{pk}, m) \rightarrow c$: Encrypts message m under public key pk .
4. $\mathcal{PKE}.\text{Dec}(\text{sk}, c) \rightarrow m$: Decrypts ciphertext c using secret key sk .

To enable explicit control over randomness for reproducible encryption inside the NP relation, we use an equivalent extended encryption syntax

$$\mathcal{PKE}.\text{Enc}(\text{pk}, m, \rho) \rightarrow c,$$

where ρ is sampled from the scheme-defined randomness space. The standard probabilistic encryption algorithm corresponds to sampling fresh randomness internally.

IND-CPA Security. For any PPT adversary $\mathcal{A}$, the IND-CPA advantage is at most $\text{negl}(\lambda)$.

Correctness. For every honestly generated key pair (sk, pk) and every message m ,

$$\mathcal{PKE}.\text{Dec}(\text{sk}, \mathcal{PKE}.\text{Enc}(\text{pk}, m, \rho)) = m$$

for all valid encryption randomness ρ .

3.3 Vector Commitments

Vector commitments (VCs) enable committing to an ordered vector of values $\mathbf{v} = (v_1, v_2, \dots, v_n)$ and later proving statements about individual positions without revealing the entire vector. A VC scheme comprises:

1. $\mathcal{VC}.\text{Setup}(1^\lambda, n) \rightarrow \text{pp}$: Generates public parameters for vectors of maximum length n .
2. $\mathcal{VC}.\text{Commit}(\mathbf{v}) \rightarrow (\text{com}, \text{st})$: Produces a commitment com to vector $\mathbf{v}$ and state st .
3. $\mathcal{VC}.\text{Prove}(\text{st}, i, v_i) \rightarrow \pi_i$: Generates a proof π_i that position i contains value v_i .
4. $\mathcal{VC}.\text{Verify}(\text{com}, i, v_i, \pi_i) \rightarrow \{0, 1\}$: Verifies the proof.

Binding. For any PPT adversary $\mathcal{A}$, the probability of producing two different vectors $\mathbf{v} \neq \mathbf{v}'$ that commit to the same com is at most $\text{negl}(\lambda)$.

Positional Binding. For any PPT adversary $\mathcal{A}$, the probability of producing a commitment com , an index i , and two distinct values $v_i \neq v'_i$ with proofs π_i, π'_i such that

$$\mathcal{VC}.\text{Verify}(\text{com}, i, v_i, \pi_i) = 1 \quad \text{and} \quad \mathcal{VC}.\text{Verify}(\text{com}, i, v'_i, \pi'_i) = 1$$

is at most $\text{negl}(\lambda)$.

3.4 Zero-Knowledge Proof Systems

Zero-knowledge proof (ZKP) systems allow a prover to convince a verifier of the truth of a statement without revealing any additional information beyond the statement's validity. We use a non-interactive zero-knowledge (NIZK) proof system with:

1. $\mathcal{ZKP}.\text{Setup}(1^\lambda) \rightarrow \text{pp}$: Generates the common reference string or other necessary public parameters.
2. $\mathcal{ZKP}.\text{Prove}(\mathcal{R}, x, w) \rightarrow \pi$: Generates a proof π for NP relation $\mathcal{R}$ with public input x and witness w .
3. $\mathcal{ZKP}.\text{Verify}(\mathcal{R}, x, \pi) \rightarrow \{0, 1\}$: Verifies the proof.
4. $\mathcal{ZKP}.\text{Sim}(\mathcal{R}, x) \rightarrow \pi$: Generates a simulated proof for true statements without using the witness.

The statement to be proven is that a given NP relation $\mathcal{R}$ holds for some witness w , i.e., $\mathcal{R}(x, w) = 1$, where x is public input and w is private witness.

Completeness. An honest prover can always convince an honest verifier of a true statement.

Soundness. For any PPT adversary $\mathcal{A}$, the probability of producing a valid proof for a false statement is at most $\text{negl}(\lambda)$.

Zero-Knowledge. For any PPT distinguisher $\mathcal{D}$, the advantage in distinguishing a real proof from a simulated proof is at most $\text{negl}(\lambda)$.

4 Security Model

In this section, we formalize the security goals of auditable stealth addresses. Our framework aims to simultaneously provide public-transcript receiver unlinkability against passive observers, receiver-whitelist compliance enforcement against malicious senders, and guarantee that authorized auditors recover receiver-linked audit information consistent with the same receiver used in stealth-address derivation.

4.1 Trust Assumptions and Scope

Our framework targets *receiver-side compliance enforcement* in amount-transparent blockchain settings. Specifically, it ensures that any accepted transaction is directed to a receiver whose public key pair belongs to a whitelist committed by the compliance authority.

We do not aim to hide the set of compliant users; rather, our privacy goal is to prevent public linkage between an accepted transaction and a specific whitelisted receiver. Sender privacy, amount privacy, source-of-funds analysis, and richer compliance predicates (e.g., amount thresholds, jurisdiction-specific rules, or purpose restrictions) are outside the scope of this work.

We assume that the compliance authority honestly maintains and publishes a canonical whitelist commitment, and that authorized auditors legitimately

hold the decryption key used for transaction-level auditability. In practice, the verifier accepts only whitelist commitments explicitly authorized on-chain by the compliance authority, such as the current commitment or a bounded set of recent epoch commitments. This prevents a sender from selecting an arbitrary, stale, or unauthorized root while still allowing operational whitelist updates.

4.2 Overall Security Objectives

Our framework targets two primary security goals and one complementary functional guarantee:

1. **Public-transcript Receiver Unlinkability:** Protect receiver privacy by ensuring that accepted transactions with stealth addresses cannot be linked to the same receiver by public observers.
2. **Compliance Soundness:** Prevent malicious senders from constructing valid transactions for non-whitelisted receivers.
3. **Audit Correctness:** Ensure that for every accepted transaction, authorized auditors decrypt receiver-linked audit information that is consistent with the same receiver used in stealth-address derivation.

4.3 Privacy Security Model (Model 1)

Security Goal¹: Ensure unlinkability of accepted stealth transactions against passive public observers.

Threat Model:

- Adversary: A passive public observer.
- Other parties: The sender and receiver are honest.
- Adversary Capabilities:
 - Can observe all public on-chain data, including transactions $(\text{addr}, \text{hint}, c, \pi_{\text{zk}})$.
 - Can locally execute $\mathcal{ZKP}.\text{Verify}(\mathcal{R}, x, \pi) \rightarrow \{0, 1\}$ to verify proof π_{zk} of observed transactions.
 - Cannot execute $\mathcal{SA}.\text{CheckAddr}$ (lacks sk_{view}).
 - Cannot decrypt ciphertext c (lacks sk_{aud}).

Definition 1 (Public-Transcript Receiver Unlinkability). *An instantiation of zkBSA satisfies public-transcript receiver unlinkability if for any PPT adversary $\mathcal{A}$, its advantage $\text{Adv}_{\text{zkBSA}, \mathcal{A}}^{\text{pt-unlink}}(\lambda)$ in the experiment $\text{Exp}_{\mathcal{A}}^{\text{pt-unlink}}(\lambda)$ is negligible.²*

¹ This goal applies to public observers other than authorized auditors. Auditors are allowed to recover receiver-linked audit information.

² Definitions of $\text{Adv}_{\text{zkBSA}, \mathcal{A}}^{\text{pt-unlink}}(\lambda)$ and $\text{Exp}_{\mathcal{A}}^{\text{pt-unlink}}(\lambda)$ are deferred to Appendix A.

4.4 Compliance Security Model (Model 2)

Security Goal: Prevent malicious senders from constructing valid transactions for non-whitelisted receivers.

Threat Model:

- Adversary: A malicious sender.
- Other parties: The verifier, compliance authority, and auditor are honest.
- Adversary Capabilities:
 - Can construct arbitrary transaction components $(\text{addr}, \text{hint}, c, \pi_{zk})$.
- Policy Knowledge:
 - The whitelist $\mathbf{W}$ is public, and the adversary’s goal is to bypass cryptographic enforcement of whitelist membership despite having full knowledge of compliant entities.

Definition 2 (Compliance Soundness). *An instantiation of zkBSA satisfies compliance soundness if no PPT adversary can produce an accepted transaction $(\text{addr}, \text{hint}, c, \pi_{zk})$ under authorized whitelist commitment com_W such that there exists no whitelisted receiver key pair $(\text{pk}_{\text{spend}}, \text{pk}_{\text{view}}) \in \mathbf{W}$ and witness $(i, \text{pk}_{\text{spend}}, \text{pk}_{\text{view}}, \pi_{vc}, r, \rho)$ satisfying relation $\mathcal{R}$.*

4.5 Audit Correctness

Definition 3 (Audit Correctness). *An instantiation of zkBSA satisfies audit correctness if for every accepted transaction $\text{tx} = (\text{addr}, \text{hint}, c, \pi_{zk})$, an authorized auditor decrypting*

$$\text{info} \leftarrow \mathcal{PK}\mathcal{E}.\text{Dec}(\text{sk}_{\text{aud}}, c)$$

recovers audit information deterministically associated with the same receiver key pair used in stealth-address derivation and whitelist membership verification.

In our concrete instantiation, we set $\text{info} = \text{pk}_{\text{spend}}$. Hence, audit correctness means that any accepted transaction decrypts to the spending public key of the same whitelisted receiver whose key pair was used to derive $(\text{addr}, \text{hint})$. Mapping this recovered public key to a real-world identity relies on the compliance authority’s off-chain registration records rather than on an additional cryptographic guarantee.

Theorem 1 (Security of zkBSA). *Assume that SA provides receiver unlinkability, $\mathcal{PK}\mathcal{E}$ satisfies IND-CPA security and correctness, $\mathcal{VC}$ satisfies positional binding, and $\mathcal{ZKP}$ satisfies soundness and zero-knowledge. Then an instantiation of zkBSA satisfies public-transcript receiver unlinkability, compliance soundness, and audit correctness.*

Proof. The result follows from Theorem 2, Theorem 3, and Theorem 4. See Appendix A.

5 zkBSA: Auditable and Compliant Stealth Address Framework

In this section, we present zkBSA—our generic framework for auditable stealth addresses that reconciles receiver privacy with compliance enforcement. Our design centers on modularity: the framework decouples privacy preservation from compliance verification, allowing independent selection of cryptographic primitives while maintaining end-to-end guarantees.

5.1 High-Level Architecture

zkBSA is designed for regulated digital assets, such as stablecoins, where the issuer acts as the compliance authority responsible for KYC and AML checks.

Our framework comprises four components:

1. **Stealth Address Scheme ($\mathcal{SA}$)**: provides receiver unlinkability and one-time address generation.
2. **Public-Key Encryption Scheme ($\mathcal{PKE}$)**: securely transmits receiver-linked audit information to authorized auditors.
3. **Vector Commitment Scheme ($\mathcal{VC}$)**: commits to compliance policies and enables efficient membership proofs.
4. **Zero-Knowledge Proof System ($\mathcal{ZKP}$)**: cryptographically binds all components while preserving privacy.

The framework operates in three phases: setup, transaction, and audit/spending.

5.2 Framework Components

Setup Phase. During setup, system parameters are initialized and compliance policies are established.

- **Global Setup:** $\text{Setup}(1^\lambda) \rightarrow \text{pp}$
 - Execute $\mathcal{SA}.\text{Setup}(1^\lambda)$, $\mathcal{PKE}.\text{Setup}(1^\lambda)$, $\mathcal{VC}.\text{Setup}(1^\lambda, n)$, and $\mathcal{ZKP}.\text{Setup}(1^\lambda)$.
 - Output global public parameters³ pp .
- **Receiver Registration:** $\text{RegisterReceiver}() \rightarrow ((\text{sk}_{\text{spend}}, \text{sk}_{\text{view}}), (\text{pk}_{\text{spend}}, \text{pk}_{\text{view}}))$
 - Each compliant receiver independently generates:
 - * $(\text{sk}_{\text{spend}}, \text{pk}_{\text{spend}}) \leftarrow \mathcal{SA}.\text{SpendKeyGen}(\text{pp}_{\mathcal{SA}})$
 - * $(\text{sk}_{\text{view}}, \text{pk}_{\text{view}}) \leftarrow \mathcal{SA}.\text{ViewKeyGen}(\text{pp}_{\mathcal{SA}})$
 - Receivers submit their public keys $(\text{pk}_{\text{spend}}, \text{pk}_{\text{view}})$ to the compliance authority for authorized whitelist inclusion.
- **Authorized Whitelist Commitment:** $\text{CommitWhitelist}(\mathbf{W}) \rightarrow (\text{com}_W, \text{st}_W)$

³ Throughout this paper, the public parameters $\text{pp} = (\text{pp}_{\mathcal{SA}}, \text{pp}_{\mathcal{PKE}}, \text{pp}_{\mathcal{VC}}, \text{pp}_{\mathcal{ZKP}})$ generated by Global Setup are treated as implicit global inputs to subsequent algorithms. For brevity, we omit them from algorithm signatures and pseudocode.

- The compliance authority constructs the authorized whitelist vector

$$\mathbf{W} = [(\text{pk}_{\text{spend},1}, \text{pk}_{\text{view},1}), \dots, (\text{pk}_{\text{spend},n}, \text{pk}_{\text{view},n})].$$

- Compute commitment $(\text{com}_W, \text{st}_W) \leftarrow \mathcal{VC}.\text{Commit}(\mathbf{W})$.
- Publish com_W on-chain as the compliance anchor and publish $\mathbf{W}$ off-chain to facilitate audit and proof generation.⁴
- **Auditor Key Initialization:** $\text{InitAuditorKeys}() \rightarrow (\text{sk}_{\text{aud}}, \text{pk}_{\text{aud}})$
 - Authorized auditors generate key pairs $(\text{sk}_{\text{aud}}, \text{pk}_{\text{aud}}) \leftarrow \mathcal{PK}\mathcal{E}.\text{KeyGen}(\text{pp}_{\mathcal{PK}\mathcal{E}})$.
 - Each auditor sends pk_{aud} to the compliance authority.
 - The compliance authority publishes pk_{aud} on-chain.

Transaction Phase When a sender wishes to transfer funds to a receiver while satisfying compliance requirements:

- **Address Derivation:** $\text{DeriveStealthAddress}((\text{pk}_{\text{spend}}, \text{pk}_{\text{view}}), r) \rightarrow (\text{addr}, \text{hint})$
 - The sender samples $r \xleftarrow{\$} \{0, 1\}^\lambda$.
 - Compute $(\text{addr}, \text{hint}) \leftarrow \mathcal{SA}.\text{DeriveAddr}((\text{pk}_{\text{spend}}, \text{pk}_{\text{view}}), r)$.
- **Audit Information Preparation:** $\text{PrepareAuditInfo}(\text{pk}_{\text{spend}}, \text{pk}_{\text{aud}}) \rightarrow c$
 - In our framework, define $\text{info} := \text{pk}_{\text{spend}}$.⁵
 - The sender samples encryption randomness $\rho \xleftarrow{\$} \mathcal{R}_{\text{enc}}$ from the scheme-defined randomness space.
 - Encrypt $c \leftarrow \mathcal{PK}\mathcal{E}.\text{Enc}(\text{pk}_{\text{aud}}, \text{info}, \rho)$.
- **Compliance Proof Generation:**

$$\text{GenerateComplianceProof}(\text{com}_W, \text{pk}_{\text{spend}}, \text{pk}_{\text{view}}, \text{addr}, \text{hint}, c, r, \rho) \rightarrow \pi_{\text{zk}}$$

- The sender computes a VC membership proof⁶

$$\pi_{\text{vc}} \leftarrow \mathcal{VC}.\text{Prove}(\text{st}_W, i, (\text{pk}_{\text{spend}}, \text{pk}_{\text{view}})),$$

where the receiver's key pair is at position i in $\mathbf{W}$.

- Construct the ZK proof $\pi_{\text{zk}} \leftarrow \mathcal{ZK}\mathcal{P}.\text{Prove}(\mathcal{R}, x, w)$, where the NP relation $\mathcal{R}(x, w) = 1$ iff

$$\begin{cases} \mathcal{VC}.\text{Verify}(\text{com}_W, i, (\text{pk}_{\text{spend}}, \text{pk}_{\text{view}}), \pi_{\text{vc}}) = 1, \\ (\text{addr}, \text{hint}) = \mathcal{SA}.\text{DeriveAddr}((\text{pk}_{\text{spend}}, \text{pk}_{\text{view}}), r), \\ \text{info} = \text{pk}_{\text{spend}}, \\ c = \mathcal{PK}\mathcal{E}.\text{Enc}(\text{pk}_{\text{aud}}, \text{info}, \rho). \end{cases}$$

⁴ When the whitelist changes, the compliance authority publishes a new authorized commitment. The verifier accepts only commitments that are currently authorized on-chain (or belong to an explicitly allowed epoch window), thereby preventing use of stale or unauthorized whitelist commitments.

⁵ info can be generalized, see Appendix A.

⁶ In the Merkle-tree instantiation, since $\mathbf{W}$ is public off-chain, the sender can reconstruct the authentication path locally; thus st_W is only a conceptual abstraction at the generic $\mathcal{VC}$ level.

- * Public input: $x = (\text{com}_W, \text{addr}, \text{hint}, c, \text{pk}_{\text{aud}})$
- * Witness: $w = (i, \text{pk}_{\text{spend}}, \text{pk}_{\text{view}}, \pi_{\text{vc}}, r, \rho)$
- **Transaction Publication:** The sender publishes the transaction $(\text{addr}, \text{hint}, c, \pi_{\text{zk}})$.

Verification, Audit, and Spending Phase

- **On-Chain Verification:** Any network participant or smart contract verifies compliance by checking both that com_W is an authorized whitelist commitment and that $\mathcal{ZKP}.\text{Verify}(\mathcal{R}, x, \pi_{\text{zk}}) = 1$. Invalid transactions are rejected immediately.
- **Auditor Decryption:** Authorized auditors decrypt $\text{info} \leftarrow \mathcal{PKE}.\text{Dec}(\text{sk}_{\text{aud}}, c)$. In our concrete instantiation, $\text{info} = \text{pk}_{\text{spend}}$. Since the ZK relation binds this value to the same whitelisted key pair used in stealth-address derivation, the auditor can reliably map each accepted transaction to its corresponding whitelisted receiver key and, via the compliance authority’s off-chain records, to the associated registered identity if needed.
- **Receiver Spending:** Upon observing a transaction with $(\text{addr}, \text{hint}, c, \pi_{\text{zk}})$, a receiver runs $\mathcal{SA}.\text{CheckAddr}(\text{sk}_{\text{view}}, \text{hint}, \text{addr})$. If the output is 1, the transaction belongs to them. The receiver then recovers the one-time spending key $\text{sk}_{\text{addr}} \leftarrow \mathcal{SA}.\text{RecoverKey}((\text{sk}_{\text{spend}}, \text{sk}_{\text{view}}), \text{hint})$ and spends the funds without revealing their long-term identity.

5.3 Instantiation Flexibility

A key advantage of zkBSA is its agnosticism to specific cryptographic implementations:

- Stealth Address Schemes: blockchain-specific
- Encryption Schemes: any IND-CPA secure PKE scheme
- Vector Commitments: Merkle trees, KZG-based [19], or other constructions
- ZK Proof Systems: SNARKs [19], STARKs [2], or zkVM-based approaches [3]

This modularity enables deployment across diverse blockchain platforms and adaptation to evolving cryptographic standards, including post-quantum migration paths.

5.4 Application Scenarios

Our framework is well suited for identity-sensitive, amount-transparent transactions that require receiver-side compliance enforcement:

- Institutional Treasury Operations: protecting counterparty identity in treasury transfers while preserving auditability and enforcing receiver-side compliance;
- Compliance-Gated Token Transfers: restricting transfers to authorized entities without exposing the recipient identity to the public;

- Compliant Exit from Regulated Privacy Pools / Mixers: in regulated privacy systems such as Zeto [10], users may later withdraw funds from private UTXOs into an account-based address for settlement, treasury management, or downstream smart-contract interactions. A direct withdrawal to a regular address reveals the destination publicly, while an unrestricted hidden destination may violate KYC or whitelist policies. zkBSA provides a compliant middle ground: assets can exit the privacy pool to a stealth address that is publicly verifiable to belong to an authorized recipient set, while keeping the exact recipient hidden from public observers and still allowing authorized auditors to recover receiver-linked audit information.

5.5 Clarification on Public Information

Our privacy guarantee is specifically designed to protect against transaction linkability. Since the whitelist $\mathbf{W}$ of compliant receivers is public, the system does reveal the set of compliant entities. However, the underlying stealth-address scheme ensures that an observer cannot link any specific accepted transaction address to a particular entry in $\mathbf{W}$. This design choice improves policy transparency and simplifies the auditor’s workflow without compromising the core privacy objective.

The public availability of hint does not compromise unlinkability. In Monero-style or ERC-5564-style stealth-address schemes, hint is typically the ephemeral public key used in an elliptic-curve Diffie–Hellman key agreement, which does not reveal useful receiver-identifying information under standard assumptions such as CDH.

6 Proof of Concept and Experimental Evaluation

This section presents a proof-of-concept (PoC) implementation of zkBSA, demonstrating its practical feasibility in a realistic blockchain setting. Our implementation is self-contained, executed locally on commodity hardware with GPU acceleration, and faithfully instantiates the modular design specified in Section 5 and Appendix A.

6.1 Implementation Overview and Technical Stack

We implement the end-to-end protocol pipeline, including receiver registration, transaction construction, and on-chain verification. All computationally intensive operations are performed off-chain; only the final compliance check is enforced on-chain via a smart contract.⁷

Table 1 lists the system components.

⁷ Readers can find our PoC open-source implementation here:

(i) Prover: <https://github.com/andrewcoder666/zkbsa-risc0-prover>
(ii) Verifier: <https://github.com/andrewcoder666/zkbsa-contracts>

Two-Stage ZK Proving with RISC Zero. Our ZK circuit is implemented as a Rust guest program that encodes the NP relation $\mathcal{R}$. RISC Zero zkVM generates proofs in two stages:

1. **STARK Phase:** a transparent proof over the RISC-V execution trace (no trusted setup).
2. **Groth16 Compression Phase:** recursive proving and compression produce a constant-size Groth16 proof over BN254 suitable for on-chain verification.

Table 1. Component instantiations

Component	Instantiation	Rationale
Stealth Address	ERC-5564 (secp256k1)	Ethereum standard; supports dual-key architecture
Public-Key Encryption	EC-ElGamal (secp256k1)	IND-CPA secure; efficient for short messages; no need to encode plaintext to curve when $\text{info} = \text{pk}_{\text{spend}}$
Vector Commitment	Merkle Tree (SHA-256)	Logarithmic proofs; minimal on-chain footprint (only 32 bytes root)
Zero-Knowledge Proof	RISC Zero zkVM (v3.0.5)	General-purpose; enables complex logic in Rust

6.2 Experimental Setup

- Hardware: Intel Core i7-13700K processor, 64 GB RAM, NVIDIA GeForce RTX 4080 GPU
- OS: Ubuntu 24.04 LTS
- Blockchain emulator: Foundry
- ZK stack: RISC Zero zkVM v3.0.5 (local GPU proving)
- Whitelist sizes: $n = |\mathbf{W}| \in \{2^4, 2^8, 2^{12}, 2^{16}, 2^{20}, 2^{24}\}$

Timing includes full I/O and proof serialization. Each data point is averaged over 10 runs.

6.3 Performance Evaluation

Scalability Analysis. Table 2 shows performance as n scales. Total proving time ranges from 3,948ms to 5,297ms.

We tested zkBSA on the Ethereum virtual machine for proof verification. Due to the constant verification time of Groth16, the on-chain proof verification costs 234,820 gas and the total transaction cost is 293,353 gas, both independent of n .

Table 2. Proving time for different whitelist sizes n

n	STARK Proving Time (ms)			Total Proving Time (ms)		
	Min	Max	Avg.	Min	Max	Avg.
2^4	1,254	1,619	1,351.9	3,948	4,276	4,125.3
2^8	1,445	1,625	1,519.8	4,775	4,935	4,844.4
2^{12}	1,641	1,790	1,708.9	4,898	5,104	5,011.9
2^{16}	1,897	1,946	1,915.0	5,124	5,269	5,222.8
2^{20}	1,902	2,105	1,933.6	5,124	5,287	5,237.6
2^{24}	1,917	2,130	1,951.1	5,161	5,297	5,255.8

Comparison Table 3 situates our approach ($n = 2^{16}$) relative to representative baselines. These schemes target different privacy goals, so the comparison is intended to highlight trade-offs rather than provide a perfectly like-for-like benchmark.

Due to different proof systems, our zkBSA implementation cannot be directly compared end-to-end with Anonymous PGC. Therefore, in this part we focus on gas costs excluding proof verification. If Anonymous PGC were migrated to a Groth16-style on-chain verifier, the proof-verification gas cost would be of the same order of magnitude.

In Anonymous PGC⁸, confidential transactions require encrypted transfer amounts and encrypted account balances. On-chain homomorphic operations (300 gas for amount addition) therefore consume significant gas, and the gas cost grows linearly with anonymity size n .

Table 3. Comparison of privacy and compliance properties across schemes

Scheme	Sender Privacy	Receiver Privacy	Amount Transparency	On-Chain Compliance	Gas Cost (without proof verification)
ERC-5564 [20]	×	✓	✓	×	~38k
Whitelist Transfer	×	×	✓	✓	~35k
Anonymous PGC [24]	×	✓	×	✓	> 19,661k
zkBSA (Ours)	×	✓	✓	✓	~59k

⁸ Anonymous PGC [24] and its similar works [4–6, 24] cannot offer sender’s privacy, because signing blockchain transactions reveals the sender identity. To hide sender’s identity, ring signatures [17] or relayers [15] are required.

6.4 Discussion of Practical Implications

Our experimental results suggest that zkBSA is practically feasible for regulated, moderate-throughput transfer scenarios:

- **Scalability:** proof generation remains under 5.3 seconds even for whitelist sizes up to 2^{24} (over 16 million users).
- **Verification Efficiency:** on-chain verification gas cost of 234,820 is compatible with blockchain validation.
- **Deployment Flexibility:** the modular design allows adaptation to different blockchain platforms and cryptographic standards.

The main trade-off is increased off-chain computation during transaction generation. This overhead is acceptable in settings where senders have more computational resources than validators and transaction generation is not latency critical. Production systems can compile the ZK relation in a specialized circuit instead of using generic zkVM to accelerate the proving phase.

7 Conclusion

In this work, we introduced zkBSA, a modular framework for auditable stealth addresses that reconciles two important goals in regulated blockchains: receiver privacy and enforceable compliance. By composing stealth addresses, public-key encryption, vector commitments, and zero-knowledge proofs, zkBSA supports transactions that are identity-sensitive, amount-transparent, and receiver-side compliance-enforced.

Our key insight is that privacy and compliance need not be treated as mutually exclusive if they are captured under separate threat models. zkBSA distinguishes a passive public observer from a malicious sender. Within this model: (i) public verifiers can reject non-compliant transactions on-chain via succinct zero-knowledge proofs; (ii) authorized auditors can recover receiver-linked audit information from accepted transactions; (iii) legitimate receivers retain unlinkability against public observers.

We provided formal definitions and proofs for public-transcript receiver unlinkability, compliance soundness, and audit correctness, showing that these guarantees reduce to standard assumptions on the underlying cryptographic components. Our proof-of-concept implementation on a realistic stack demonstrates feasibility at practical scales, supporting registries of over 16 million users with fixed on-chain verification cost.

Beyond direct payments, zkBSA is particularly useful as a compliant destination mechanism for assets leaving regulated privacy systems. In workflows where private UTXOs are later consolidated into an account-based environment, zkBSA can preserve recipient privacy at the exit boundary while maintaining whitelist-constrained receiver compliance and auditability.

Looking forward, zkBSA’s modular architecture is promising for future extensions, including richer compliance predicates, dynamic whitelist updates, multi-auditor settings, and post-quantum instantiations.

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A Formal Security Analysis and Discussion

A.1 Public-Transcript Receiver Unlinkability Proof

Experiment $\text{Exp}_{\mathcal{A}}^{\text{pt-unlink}}(\lambda)$:

1. The challenger runs the setup algorithms and gives $\mathcal{A}$ all public parameters, the authorized whitelist commitment com_W , the public whitelist $\mathbf{W}$, and the auditor public key pk_{aud} .

2. $\mathcal{A}$ outputs two distinct challenge receivers

$$R_0 = (\mathbf{pk}_{\text{spend},0}, \mathbf{pk}_{\text{view},0}), \quad R_1 = (\mathbf{pk}_{\text{spend},1}, \mathbf{pk}_{\text{view},1}),$$

such that both $R_0, R_1 \in \mathbf{W}$.

3. The challenger samples a random bit $b \xleftarrow{\$} \{0,1\}$, sender randomness r , and encryption randomness ρ , and generates an honest accepted public transaction transcript for receiver R_b :

$$(\text{addr}, \text{hint}) \leftarrow \mathcal{SA}.\text{DeriveAddr}(R_b, r), \quad (1)$$

$$c \leftarrow \mathcal{PKE}.\text{Enc}(\mathbf{pk}_{\text{aud}}, \mathbf{pk}_{\text{spend},b}, \rho), \quad (2)$$

$$\pi_{\text{zk}} \leftarrow \mathcal{ZKP}.\text{Prove}(\mathcal{R}, x, w) \quad (3)$$

where $x = (\text{com}_W, \text{addr}, \text{hint}, c, \mathbf{pk}_{\text{aud}})$ and w is the corresponding valid witness.

4. The challenger gives the complete accepted public transcript $\text{tx}_{\text{pub}} = (\text{addr}, \text{hint}, c, \pi_{\text{zk}})$ to $\mathcal{A}$.
5. $\mathcal{A}$ outputs a guess bit b' .

Define the advantage as $\text{Adv}_{\text{zkBSA}, \mathcal{A}}^{\text{pt-unlink}}(\lambda) = |\Pr[b' = b] - \frac{1}{2}|$.

Theorem 2 (Public-Transcript Receiver Unlinkability of zkBSA). *Assume that the underlying stealth-address scheme $\mathcal{SA}$ satisfies receiver unlinkability, the public-key encryption scheme $\mathcal{PKE}$ satisfies IND-CPA security, and the zero-knowledge proof system $\mathcal{ZKP}$ satisfies zero-knowledge for relation $\mathcal{R}$. Then zkBSA satisfies public-transcript receiver unlinkability.*

Proof. Fix any PPT adversary $\mathcal{A}$ in $\text{Exp}_{\mathcal{A}}^{\text{pt-unlink}}(\lambda)$. For $b \in \{0,1\}$, let

$$T_b = (\text{addr}_b, \text{hint}_b, c_b, \pi_b)$$

denote the distribution of the complete accepted public transaction transcript honestly generated for challenge receiver $R_b = (\mathbf{pk}_{\text{spend},b}, \mathbf{pk}_{\text{view},b})$. We must show that no PPT adversary can distinguish T_0 from T_1 with non-negligible advantage.

Suppose, towards contradiction, that some PPT adversary $\mathcal{A}$ distinguishes T_0 and T_1 with non-negligible advantage. Since each transcript is honestly generated and accepted, the proof component π_b is a proof for a true statement

$$x_b = (\text{com}_W, \text{addr}_b, \text{hint}_b, c_b, \mathbf{pk}_{\text{aud}}).$$

By the zero-knowledge property of $\mathcal{ZKP}$, the proof π_b reveals no information beyond the public statement x_b . Therefore, any non-negligible distinguishing advantage of $\mathcal{A}$ must come from the difference between the public statements

$$x_0 = (\text{com}_W, \text{addr}_0, \text{hint}_0, c_0, \mathbf{pk}_{\text{aud}}) \quad \text{and} \quad x_1 = (\text{com}_W, \text{addr}_1, \text{hint}_1, c_1, \mathbf{pk}_{\text{aud}}).$$

The components com_W and $\mathbf{pk}_{\text{aud}}$ are identical in both distributions, so the only receiver-dependent public components are $(\text{addr}_b, \text{hint}_b)$ and c_b .

Now consider the stealth-address part. If a PPT adversary could distinguish whether $(\text{addr}, \text{hint})$ was honestly derived from R_0 or from R_1 with non-negligible advantage, then this adversary could be used directly to break the receiver unlinkability of the underlying stealth-address scheme $\mathcal{SA}$. Hence the distributions $(\text{addr}_0, \text{hint}_0)$ and $(\text{addr}_1, \text{hint}_1)$ are computationally indistinguishable.

Next consider the ciphertext part. The ciphertexts are distributed as

$$c_0 \leftarrow \mathcal{PKE}.\text{Enc}(\text{pk}_{\text{aud}}, \text{pk}_{\text{spend},0}, \rho), \quad c_1 \leftarrow \mathcal{PKE}.\text{Enc}(\text{pk}_{\text{aud}}, \text{pk}_{\text{spend},1}, \rho).$$

If a PPT adversary could distinguish c_0 from c_1 with non-negligible advantage, then it would directly yield an IND-CPA adversary against $\mathcal{PKE}$ using challenge messages $m_0 = \text{pk}_{\text{spend},0}$, $m_1 = \text{pk}_{\text{spend},1}$. Therefore c_0 and c_1 are computationally indistinguishable.

It follows that the public statements x_0 and x_1 are computationally indistinguishable. Since zero-knowledge implies that the proof components π_0 and π_1 reveal no additional information beyond these statements, the complete transcripts $T_0 = (\text{addr}_0, \text{hint}_0, c_0, \pi_0)$ and $T_1 = (\text{addr}_1, \text{hint}_1, c_1, \pi_1)$ are also computationally indistinguishable. This contradicts the assumption that $\mathcal{A}$ distinguishes them with non-negligible advantage.

Hence, for every PPT adversary $\mathcal{A}$, $\text{Adv}_{\text{zkBSA},\mathcal{A}}^{\text{pt-unlink}}(\lambda) \leq \text{negl}(\lambda)$, which proves the theorem.

A.2 Compliance Soundness Proof

Theorem 3 (Compliance Soundness). *Assume the vector commitment scheme $\mathcal{VC}$ is positionally binding and the zero-knowledge proof system $\mathcal{ZKP}$ is sound. Then an instantiation of zkBSA satisfies compliance soundness against any PPT adversary $\mathcal{A}$ with probability at most $\text{negl}(\lambda)$.*

Proof. Suppose, for contradiction, that there exists a PPT adversary $\mathcal{A}$ that produces a valid transaction $\text{tx} = (\text{addr}, \text{hint}, c, \pi_{\text{zk}})$ such that the public verifier accepts but the receiver bound to the transaction is not in the whitelist $\mathbf{W}$.

Acceptance means $\mathcal{ZKP}.\text{Verify}(\mathcal{R}, x, \pi_{\text{zk}}) = 1$ for $x = (\text{com}_W, \text{addr}, \text{hint}, c, \text{pk}_{\text{aud}})$.

By soundness of $\mathcal{ZKP}$, except with negligible probability, there exists a witness $w = (i, \text{pk}_{\text{spend}}, \text{pk}_{\text{view}}, \pi_{\text{vc}}, r, \rho)$ such that $\mathcal{R}(x, w) = 1$. Hence the following all hold:

$$\mathcal{VC}.\text{Verify}(\text{com}_W, i, (\text{pk}_{\text{spend}}, \text{pk}_{\text{view}}), \pi_{\text{vc}}) = 1, \quad (4)$$

$$(\text{addr}, \text{hint}) = \mathcal{SA}.\text{DeriveAddr}((\text{pk}_{\text{spend}}, \text{pk}_{\text{view}}), r), \quad (5)$$

$$c = \mathcal{PKE}.\text{Enc}(\text{pk}_{\text{aud}}, \text{pk}_{\text{spend}}, \rho). \quad (6)$$

Since com_W is a commitment to the whitelist vector $\mathbf{W}$, and $\mathcal{VC}.\text{Verify}(\cdot) = 1$, positional binding implies that the value opened at position i is uniquely determined except with negligible probability. Hence $(\text{pk}_{\text{spend}}, \text{pk}_{\text{view}})$ is exactly the value committed at position i in $\mathbf{W}$, and therefore belongs to $\mathbf{W}$. Therefore, $(\text{pk}_{\text{spend}}, \text{pk}_{\text{view}}) \in \mathbf{W}$, contradicting the assumption that the accepted transaction is bound to a non-whitelisted receiver.

Therefore, $\Pr[\text{BadEvent}] \leq \text{Adv}_{\mathcal{ZKP}}^{\text{sound}}(\lambda) + \text{Adv}_{\mathcal{VC}}^{\text{pos-bind}}(\lambda) \leq \text{negl}(\lambda)$, which proves compliance soundness.

A.3 Audit Correctness Proof

Theorem 4 (Audit Correctness). *Assume the public-key encryption scheme $\mathcal{PKE}$ is correct, the zero-knowledge proof system $\mathcal{ZKP}$ is sound, and the vector commitment scheme $\mathcal{VC}$ is positionally binding. Then an instantiation of zkBSA satisfies audit correctness.*

Proof. Consider any accepted transaction $\text{tx} = (\text{addr}, \text{hint}, c, \pi_{\text{zk}})$. Since the public verifier accepts, we have $\mathcal{ZKP}.\text{Verify}(\mathcal{R}, x, \pi_{\text{zk}}) = 1$ for $x = (\text{com}_W, \text{addr}, \text{hint}, c, \text{pk}_{\text{aud}})$.

By soundness of $\mathcal{ZKP}$, there exists a witness $w = (i, \text{pk}_{\text{spend}}, \text{pk}_{\text{view}}, \pi_{\text{vc}}, r, \rho)$, such that:

$$\mathcal{VC}.\text{Verify}(\text{com}_W, i, (\text{pk}_{\text{spend}}, \text{pk}_{\text{view}}), \pi_{\text{vc}}) = 1, \quad (7)$$

$$(\text{addr}, \text{hint}) = \mathcal{SA}.\text{DeriveAddr}((\text{pk}_{\text{spend}}, \text{pk}_{\text{view}}), r), \quad (8)$$

$$c = \mathcal{PKE}.\text{Enc}(\text{pk}_{\text{aud}}, \text{pk}_{\text{spend}}, \rho). \quad (9)$$

By correctness of $\mathcal{PKE}$, an authorized auditor decrypting c obtains

$$\mathcal{PKE}.\text{Dec}(\text{sk}_{\text{aud}}, c) = \text{pk}_{\text{spend}}.$$

Therefore, the decrypted audit information is the spending public key of the same receiver key pair that (i) appears in the whitelist proof and (ii) is used to derive $(\text{addr}, \text{hint})$. Mapping that recovered public key to a real-world identity is then performed using the compliance authority's off-chain registration records. This proves audit correctness.

A.4 Remarks on Audit Information

In our PoC, we set

$$\text{info} = \text{pk}_{\text{spend}}.$$

This choice makes auditing straightforward: once the auditor decrypts info , the receiver can be mapped to the corresponding whitelist entry and associated off-chain identity records.

More generally, one may define

$$\text{info} = f(i, \text{pk}_{\text{spend}}, \text{pk}_{\text{view}})$$

for any deterministic encoding that enables reliable receiver recovery by the auditor. The essential requirement is that info be explicitly bound in the NP relation to the same receiver key pair used in whitelist membership verification and stealth-address derivation.